Uncertainty and Probabilistic Reasoning in AI

Dr Aishwarya S., Assistant Professor of Mathematics, MIT, Manipal

Uncertainty in AI

AI systems often face incomplete, noisy, or ambiguous information.

Sources of uncertainty:

- Sensor errors (robot vision, speech recognition)
- Incomplete knowledge (medical diagnosis)
- Unpredictable environments (self-driving cars)

Deterministic logic alone is insufficient.

→ Need for probabilistic reasoning.

Probabilistic Reasoning

Probability theory provides a framework for handling uncertainty.

Key concepts:

- Prior probability (belief before evidence)
- Conditional probability (P(A|B))
- Bayes' theorem (updating beliefs with evidence)

Example: Medical diagnosis

Prior: Disease prevalence

Evidence: Test results

Posterior: Probability patient has disease given test outcome

Applications of Probabilistic Reasoning

Natural Language Processing (speech recognition, translation)

Robotics (localization, mapping)

Computer Vision (object recognition under noise)

Decision making under uncertainty (autonomous vehicles, finance)

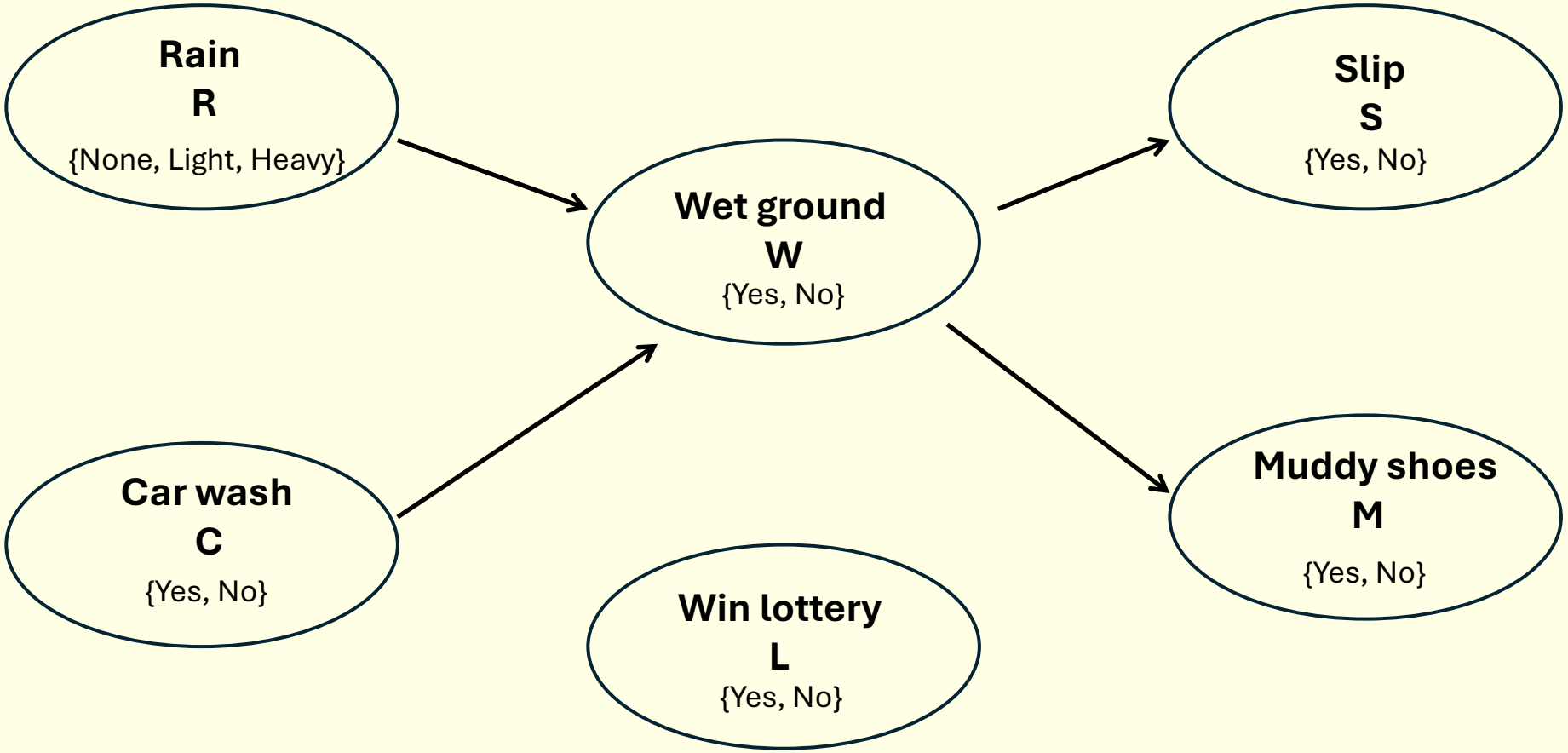
Bayesian Network

A data structure that represents the dependencies among random variables.

Bayesian Network

- A Directed Acyclic Graph (DAG).
- Each node corresponds to a random variable.
- If there is a directed edge from node X to node Y, then X is said to be a *parent* of Y.
- Each node X has associated probability distribution $P(X|Parents(X))$.

Example:


Rain
R
{None, Light, Heavy}
Slip
S
{Yes, No}
Wet ground
W
{Yes, No}
Car wash
C
{Yes, No}
Muddy shoes
M
{Yes, No}
Win lottery
L
{Yes, No}

Prior Probabilities

Rain R	P(R)
None	0.60
Light	0.30
Heavy	0.10

Car wash C	P(C)
Yes	0.10
No	0.90

Win Lottery L	P(L)
Yes	0.000001
No	0.999999

Conditional Probability Tables (CPTs)

Wet ground (W) given Rain, Car wash

Rain	Car wash	P(W=Yes)	P(W=No)
None	No	0.02	0.98
None	Yes	0.60	0.40
Light	No	0.60	0.40
Light	Yes	0.85	0.15
Heavy	No	0.95	0.05
Heavy	Yes	0.98	0.02

Slip (S) given Wet ground

Wet ground	P(S=Yes)	P(S=No)
Yes	0.15	0.85
No	0.01	0.99

Muddy shoes (M) given Wet ground

Wet ground	P(M=Yes)	P(M=No)
Yes	0.70	0.30
No	0.05	0.95

Computing Joint Probabilities

The joint probability distribution $P(X_1 = x_1,\ X_2 = x_2, \dots, X_n = x_n)$ is written shortly as $P(x_1, x_2, \dots, x_n)$.

$$
\begin{aligned}
P(x_1, x_2, \dots, x_n) &= P(x_n |\, x_{n-1}, x_{n-2}, \dots, x_1) \cdot P(x_{n-1}, x_{n-2}, \dots, x_1) \\
&= P(x_n |\, x_{n-1}, x_{n-2}, \dots, x_1) \cdot P(x_{n-1} |\, x_{n-2}, \dots, x_1) \cdot P(x_{n-2}, \dots, x_1) \\
&\ \ \vdots \\
&= P(x_n |\, x_{n-1}, x_{n-2}, \dots, x_1) \cdot P(x_{n-1} |\, x_{n-2}, \dots, x_1) \cdots P(x_2 | x_1) \cdot P(x_1).
\end{aligned}
$$

Hence, $P(x_1, x_2, \dots, x_n) = \prod_{i=1}^{n} P(x_i | x_1, x_2, \dots, x_{i-1})$.

In a Bayesian network,

$$P(x_i|x_1, x_2, \dots, x_{i-1}) = \prod_{i=1}^{n} P(x_i|Parents(X_i)),$$

assuming a labelling such that $Parents(X_i) \subseteq \{X_1, X_2, \dots, X_{i-1}\}$.

Hence,

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^{n} P(x_i|Parents(X_i)).$$

Example:

In the earlier Bayesian network, we have

$$P(R, C, W, S, M, L) = P(R)P(C)P(L)P(W|R, C)P(S|W)P(M|W).$$

For example,

$$\begin{aligned} &P(R = Light, C = No, W = Yes, S = Yes, M = Yes, L = No) \\ &\quad = P(R = Light)\, P(C = No)\, P(L = No)\, P(W = Yes|R = Light, C = No)\, P(S = Yes|W = Yes)\, P(M = Yes|W = Yes) \\ &\quad = 0.30 \times 0.90 \times 0.999999 \times 0.60 \times 0.15 \times 0.70 \\ &\quad = 0.017 \end{aligned}$$

Inference by Enumeration

$$P(X \mid e) = \alpha\, P(X, e) = \alpha \sum_y P(X, e, y)$$

X is the query variable.
e is the evidence.
y ranges over values of hidden variables.
α normalizes the result.

Example: Compute P(R | S=*yes*, M=*no*) using enumeration method.

$$P(R=r \mid S=yes, M=no)$$

$$= \alpha \; P(R=r, S=yes, M=no)$$

$$= \alpha \sum_{l} \sum_{c} \sum_{w} P(R=r, S=yes, M=no, L=l, C=c, W=w).$$

$$= \alpha \sum_{l} \sum_{c} \sum_{w} P(R=r)\, P(L=l)\, P(C=c)\, P(W=w \mid r, c)\, P(S=yes \mid w)\, P(M=no \mid w).$$

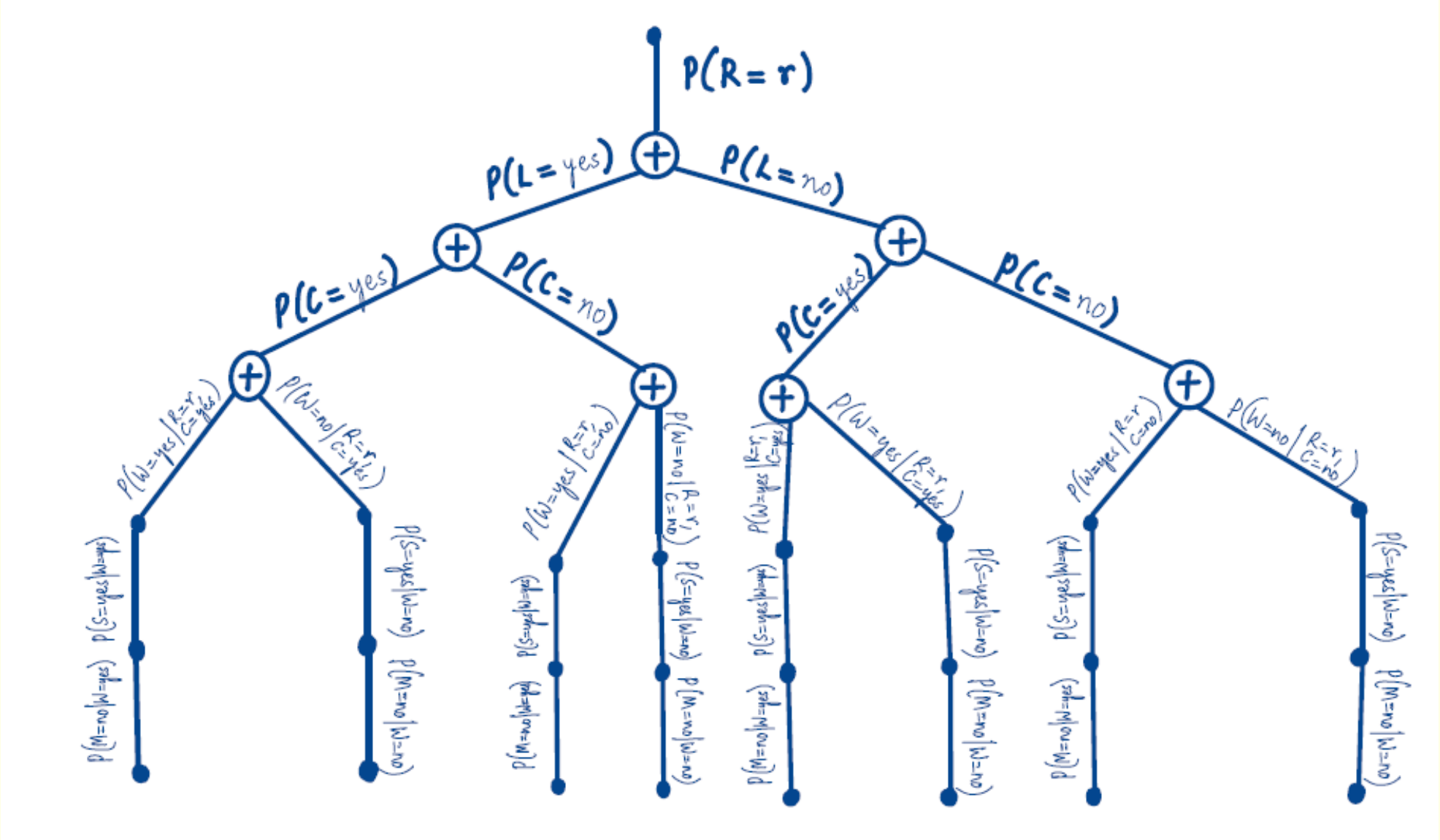
$$= \alpha\, P(R=r) \sum_{l} \left[P(L=l) \sum_{c} \left[P(C=c) \sum_{w} \left(P(W=w \mid r, c)\, P(S=yes \mid w)\, P(M=no \mid w) \right) \right] \right]$$

$$P(R = \text{none} \mid S = \text{yes}, M = \text{no})$$

$$= \alpha\,(0.6)\,[0.000001\,[0.1\,(0.6 \times 0.15 \times 0.3 + 0.4 \times 0.01 \times 0.95) + 0.9\,(0.02 \times 0.15 \times 0.3 + 0.98 \times 0.01 \times 0.95)] + 0.999999\,[0.1\,(0.6 \times 0.15 \times 0.3 + 0.4 \times 0.01 \times 0.95) + 0.9\,(0.02 \times 0.15 \times 0.3 + 0.98 \times 0.01 \times 0.95)]]$$

$$\approx \alpha\,(0.0073614).$$

Computation tree for the above example:



Similar computations for R = *light* and R = *heavy* give:
P(R=*light* | S=*yes*, M=*no*) ≈ α(0.00950637)
P(R=*heavy* | S=*yes*, M=*no*) ≈ α(0.00433315).

Hence,
P(R | S=*yes*, M=*no*) ≈ α <0.0073614, 0.00950637, 0.00433315>.

After normalizing (that is, multiplying by
$\alpha = \frac{1}{0.0073614 + 0.00950637 + 0.00433315} \approx 47.1678$),

we get the following probability distribution:

P(R | S=*yes*, M=*no*) ≈ <0.3472, 0.4484, 0.2044>.

The enumeration algorithm can be improved substantially by eliminating repeated calculations (of the type shown in the computation tree).

Inference by Variable Elimination

Key Idea:

Do the calculation once and save the results for later use.
(This is a form of Dynamic Programming.)

Example: Compute P(R | S=*yes*, M=*no*) in the above example using variable elimination method.

$$P(R=r \mid S=yes, M=no)$$

$$= \alpha \; P(R=r, S=yes, M=no)$$

$$= \alpha \sum_{l} \sum_{c} \sum_{w} P(R=r, S=yes, M=no, L=l, C=c, W=w).$$

$$= \alpha \sum_{l} \sum_{c} \sum_{w} P(R=r)\, P(L=l)\, P(C=c)\, P(W=w \mid r, c)$$
$$P(S=yes \mid w)\, P(M=no \mid w).$$

$$P(R=r \mid S=yes, M=no)$$

$$= \propto \overbrace{P(R=r)}^{f_1(R)} \sum_l \left[\overbrace{P(L=l)}^{f_2(L)} \sum_c \left[\overbrace{P(C=c)}^{f_3(C)} \sum_w \left(\overbrace{P(W=w \mid r,c)}^{f_4(W,R,C)} \overbrace{P(S=yes \mid w)}^{f_5(W)} \underbrace{P(M=no \mid w)}_{f_6(W)} \right) \right] \right]$$

where

$$f_1(R) = \begin{bmatrix} P(R=none) \\ P(R=light) \\ P(R=heavy) \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.3 \\ 0.1 \end{bmatrix}_{3\times 1}$$

$$f_2(L) = \begin{bmatrix} P(L = yes) \\ P(L = no) \end{bmatrix} = \begin{bmatrix} 0.000001 \\ 0.999999 \end{bmatrix}_{2\times 1}$$

$$f_3(C) = \begin{bmatrix} P(C = yes) \\ P(C = no) \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.9 \end{bmatrix}_{2\times 1}$$

$$f_4(W,R,C) = \left[\begin{array}{ccc} P(W=\text{yes}\mid R=\text{none},\ C=\text{yes}) & P(W=\text{yes}\mid R=\text{light},\ C=\text{yes}) & P(W=\text{yes}\mid R=\text{heavy},\ C=\text{yes}) \\ P(W=\text{yes}\mid R=\text{none},\ C=\text{no}) & P(W=\text{yes}\mid R=\text{light},\ C=\text{no}) & P(W=\text{yes}\mid R=\text{heavy},\ C=\text{no}) \\ \hline P(W=\text{no}\mid R=\text{none},\ C=\text{yes}) & P(W=\text{no}\mid R=\text{light},\ C=\text{yes}) & P(W=\text{no}\mid R=\text{heavy},\ C=\text{yes}) \\ P(W=\text{no}\mid R=\text{none},\ C=\text{no}) & P(W=\text{no}\mid R=\text{light},\ C=\text{no}) & P(W=\text{no}\mid R=\text{heavy},\ C=\text{no}) \end{array}\right]$$

$$= \left[\begin{array}{ccc} 0.6 & 0.85 & 0.98 \\ 0.02 & 0.6 & 0.95 \\ \hline 0.4 & 0.15 & 0.02 \\ 0.98 & 0.4 & 0.05 \end{array}\right]_{2\times 2\times 3}$$

(This is a 3D-array.)

$$f_5(W) = \begin{bmatrix} P(S=yes \mid W=yes) \\ P(S=yes \mid W=no) \end{bmatrix} = \begin{bmatrix} 0.15 \\ 0.01 \end{bmatrix}_{2\times 1}$$

$$f_6(W) = \begin{bmatrix} P(M=no \mid W=yes) \\ P(M=no \mid W=no) \end{bmatrix} = \begin{bmatrix} 0.3 \\ 0.95 \end{bmatrix}_{2\times 1}$$

The computations are performed by a pointwise multiplication of the above multi-dimensional arrays. This pointwise multiplication is denoted by 'x'.

This pointwise multiplication is also known as **Hadamard product**.

$$f_5(w) \times f_6(w) = \begin{bmatrix} 0.15 \\ 0.01 \end{bmatrix} \times \begin{bmatrix} 0.3 \\ 0.95 \end{bmatrix} = \begin{bmatrix} 0.045 \\ 0.0095 \end{bmatrix}$$

$$f_4(W,R,C) \times \overbrace{f_5(W) \times f_6(W)}^{\text{(computed above)}}$$

$$= \begin{bmatrix} (0.6)(0.045) & (0.85)(0.045) & (0.98)(0.045) \\ (0.02)(0.045) & (0.6)(0.045) & (0.95)(0.045) \\ \hline (0.4)(0.0095) & (0.15)(0.0095) & (0.02)(0.0095) \\ (0.98)(0.0095) & (0.4)(0.0095) & (0.05)(0.0095) \end{bmatrix}$$

$$= \begin{bmatrix} 0.027 & 0.03825 & 0.0441 \\ 0.0009 & 0.027 & 0.04275 \\ \hline 0.0038 & 0.001425 & 0.00019 \\ 0.00931 & 0.0038 & 0.000475 \end{bmatrix}$$

Summing out W :

$$f_7(R,C) = \begin{bmatrix} 0.027 & 0.03825 & 0.0441 \\ 0.0009 & 0.027 & 0.04275 \end{bmatrix} + \begin{bmatrix} 0.0038 & 0.001425 & 0.00019 \\ 0.00931 & 0.0038 & 0.000475 \end{bmatrix}$$

$$= \begin{bmatrix} 0.0308 & 0.039675 & 0.04429 \\ 0.01021 & 0.0308 & 0.043225 \end{bmatrix}$$

$$f_3(C) \times f_7(R,C) = \begin{bmatrix} 0.1 \\ 0.9 \end{bmatrix} \times \begin{bmatrix} 0.0308 & 0.039675 & 0.04429 \\ 0.01021 & 0.0308 & 0.043225 \end{bmatrix}$$

$$= \begin{bmatrix} 0.00308 & 0.0039675 & 0.004429 \\ 0.009189 & 0.02772 & 0.0389025 \end{bmatrix}$$

Summing out C :

$$f_8(R) = [0.00308 \quad 0.0039675 \quad 0.004429] + [0.009189 \quad 0.02772 \quad 0.0389025]$$

$$= [0.012269 \quad 0.0316875 \quad 0.0433315]$$

$$P(R \mid S = yes, M = no) = \alpha \cdot f_1(R) \times f_8(R)$$

$$= \alpha \cdot \begin{bmatrix} 0.6 \\ 0.3 \\ 0.1 \end{bmatrix} \times \begin{bmatrix} 0.012269 \\ 0.0316875 \\ 0.0433315 \end{bmatrix}$$

$$= \alpha \cdot \begin{bmatrix} 0.0073614 \\ 0.00950637 \\ 0.00433315 \end{bmatrix}$$

After normalizing, we get the probability distribution as below:

$$P(R \mid S = \text{yes}, M = \text{no}) = \begin{bmatrix} 0.3472 \\ 0.4484 \\ 0.2044 \end{bmatrix}.$$